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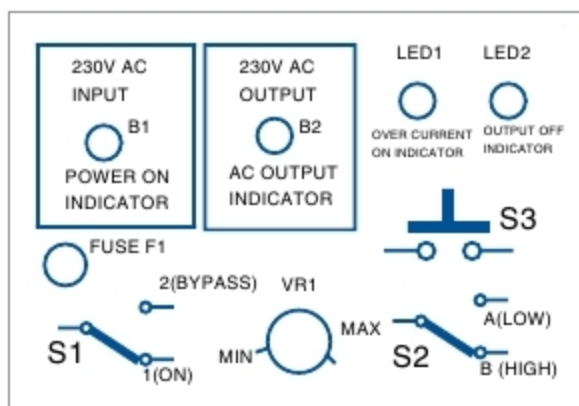


Fig. 7: Proposed panel layout



Fig. 8: Author's prototype

PARTS LIST

Semiconductors:

IC1	- 741 op-amp
T1, T2	- BC548 npn transistor
D1-D6	- 1N4007 rectifier diode
ZD1	- 15V Zener diode
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R11	- 100-kilo-ohm
R2	- 4.7-kilo-ohm
R3	- 55-kilo-ohm
R4	- 15-kilo-ohm
R5	- 1-kilo-ohm
R6, R9	- 2.2-kilo-ohm
R7, R8, R10	- 10-kilo-ohm
R12	- 100-ohm
VR1	- 5-kilo-ohm potmeter

Capacitors:

C1, C2	- 4.7 μ F, 35v electrolytic
C3	- 1000 μ F, 35V electrolytic

Miscellaneous:

CON1-CON3	- 2-pin terminal connector
B1, B2	- 230V AC neon bulb
S1	- 5A SPDT switch
S2	- SPDT switch
S3	- Tactile switch
RL1	- 12V, 1CO relay
F1	- 300mA-5A fuse with holder
X1	- 230V AC primary to 9V-0-9V, 1A secondary transformer
	- 20SWG enamelled copper wire (about 30.5cm, or 1-foot)
	- 12V regulated power supply
	- 230V AC mains power supply
	- AC load say for 230V AC, 100W bulb

included in the PCB. Connect these through external wires as per the current requirement of the load (device). Connect 230V AC across CON1 and load across CON3. Proposed front panel layout is shown in Fig. 7.

After assembling the circuit and before enclosing it in the cabinet, calibrate it as given below. Use a multimeter for measuring voltages and adjusting cutoff currents.

Calibration

1. Keep S2 either at point A or B depending on load. For low current between 300mA and 1A, keep it at point A. For high current between 1A and 5A, keep it at point B.

2. Keep current-adjustable VR1 to maximum anti-clockwise direction.

3. Connect to 230V AC input at CON1 and load (device) across CON3. Keep S1 at point 1 to enable the circuit breaker.

4. Use a multimeter to measure voltage at pin 3 of IC1.

5. Using VR1, set voltage close to the voltage of pin 3 of IC1.

6. Slowly rotate VR1 from maximum to minimum (clockwise) until LED1 glows and load gets turned off.

7. Slowly turn VR1 slightly in anti-clockwise direction.

The circuit is now ready to use.

The author's prototype is shown in Fig. 8. **EFY**



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